MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPE 4B31

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Diode, rectifier, high-vacuum

See figure 1

Mounting position: Any

Weight: 6.75 ounces (191.4 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epx	ib	Ib	tk
Unit:	Vac	kv	ma	mAdc	sec
Maximum:	5.5	18	470	150	---
Minimum:	4.5	---	---	---	120
TEST CONDITIONS:	5.0	---	---	---	120

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
1301	Heater current	-		0.65	II	If	4.6	5.4	Aac
1353	Operation of rectifiers	1, 2, 4		0.65	II	---	---	---	---
1256	Electrode current (anode)	-	Eb = 130 Vdc	0.65	II	Ib	300	---	mAdc
1231	Peak emission	3	eb = 2,500 v	0.65	II	is	18	---	a
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltage applied	---	---	---	---	---	---
1036	Bump	-	Angle = 20°	---	---	---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---	---	---
1111	Base pin solder depth	-		---	---	---	---	---	---
1105	Permanence of marking	-		---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	1	Group C: operation of rectifiers: t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission	-		---	---	is	10	---	a
---	Failure to rectify	-		---	---	---	---	---	---

NOTES:

1. Operation in rectifier circuit with resistance load and no filter at epx = 18 kv; ib = 470 ma; Ib = 150 mAdc.
2. The anode potential shall be snapped on (1-minute) and off (2-minutes) at least six times with a switch or relay. The use of a variac to apply anode potential gradually shall not be allowed.
3. When only a minimum current limit is specified, a voltage less than the value specified may be applied provided the required minimum emission current is obtained in this manner.
4. This test is to be the first test performed at the conclusion of the holding period.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU
Air Force - 80
DSA - ES

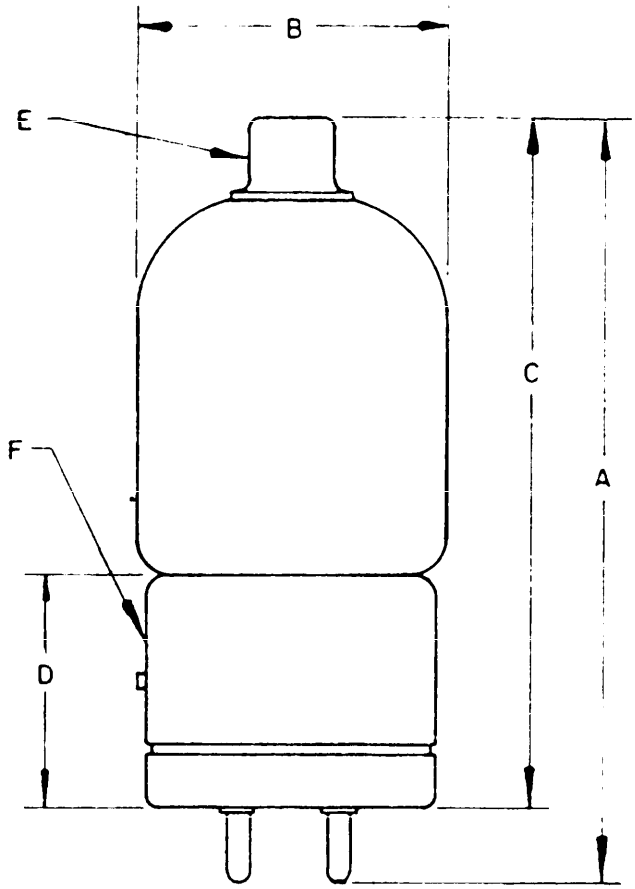
User activities:
Army - AV, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3033)

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Pin connections	
Pin No.	Element
1	connected to Pin 2 int.
2	h & k
3	h
4	connected to Pin 3 int.
cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.00(152.40)	7.00(177.80)
B		2.56 (65.02)
C	5.31(134.87)	6.31(160.27)
Quality conformance inspection, part 3 (see note)		
E	Cap: C1-5 (EIA)	
F	Base: A4-18 (except for metal sleeve length) (EIA)	
Reference dimension		
D	1.94(49.28)	

NOTE: Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 4B31.

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DEPARTMENT OF THE NAVY
NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT
DOD 316

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COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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